Bioinformatics algorithms

Rosario M. Piro
Politecnico di Milano

Variant calling – exercises

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1. **Pileup**

   (Note: knowledge about the pileup format is not relevant for the exam!)

   Given the following line from a pileup:

   ```
   1    879317    C    21    .,tTtt,.,.t.,.,.,t,T.,    C@EADE:EAHDH4>G.G93:8
   ```

   Remark (not relevant for the exercises) using the Human reference genome hg19, this is the 1830th coding base of one of the the coding transcripts of the gene SAMD11 (Sterile Alpha Motif Domain Containing 11; transcript ID: `NM_152486.3`).

   (a) How many reads matched the alternate base at this position?

   (b) How many alternate bases were sequenced on reads mapping to the reverse strand?

   (c) How many reads mapped to the reverse strand?

   (d) How many forward strand reads matched the reference sequence at this position?

   (e) What base change is this? What genotype do you think it is (homozygous reference, heterozygous, homozygous alternate)?

2. **Bayes' theorem**

   (Note: Bayes' theorem is very useful for many different applications of bioinformatics, but it is less relevant for our exam :-)

   Bayes' theorem is often used for a set of n mutually exclusive events $E_1, E_2, \ldots, E_n$ such that $\sum_i P(E_i) = 1$, and can then be written as

   $$P(E_i|B) = \frac{P(B|E_i)P(E_i)}{P(B)}$$

   An example that is seen in many textbooks describes the situation in which a chest X-ray (CXR) is obtained as a part of a routine checkup and a shadow is seen in one of the lungs that is suspicious for lung cancer. The referring physician consults a study that documents the following conditional probabilities:

   - P(CXR=positive|Lung cancer=present)=0.8
   - P(CXR=positive|Lung cancer=absent)=0.02

   That is, the CXR has a false-negative rate of 20%, and a false-positive rate of 2%. Assume for now that the prior probability of a patient having lung cancer is 1 in 1000. That is, P(Lung cancer=present)=0.001 and P(Lung cancer absent=0.999).

   (a) If we don't know $P(B)$, how can we compute it using $P(B|E_i)$? What other term(s) do we need to know?

   (b) Using your result from (a) in Eq. 2, calculate P(Lung cancer=present|CXR=positive), i.e., the posterior probability for the diagnosis of lung cancer in a person without any other symptoms who presents with a positive chest X-Ray.

3. **MAQ**

(Note: there will also be no long exercise on MAQ in the exam ...)

Given the following line from a pileup (alignment D at a specific base):

```
21   31692310   G   15   ,.,AAA.,,aaa,,A   3GFDIIIIIBIIBA+
```

and the following mapping qualities associated to the reads (in the same order):

`33, 255, 255, 255, 255, 33, 78, 255, 255, 255, 255, 255, 255, 1, 255`

The base call qualities Q_B are encoded in the string "`3GFDIIIIIBIIBA+`" (also in the same order as the bases reported in the 5th column). Although it would be interesting to learn the details about this encoding, for the purpose of this exercise you can simple translate the base qualities with the help of the following small table:

ASCII encoding	Q_B (Phred score)
`+`	10
`3`	18
`A`	32
`B`	33
`D`	35
`F`	37
`G`	38
`I`	40

Let us assume the prior probability of observing a heterozygous genotype to be $r = 0.2$ (can be used for known SNPs).

Let us further make the simplifying assumptions that the probability of observing this alignment depends linearly on the single base error proabilities, i.e.

$$\theta^i = 1 \quad \forall i,$$

and that

$$c'_{n,k} = 1 \quad \forall n, k.$$

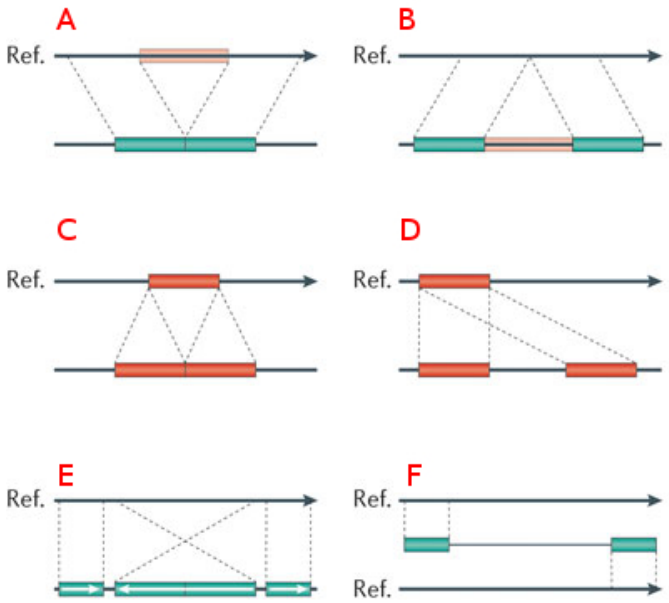
(a) Determine for all three genotypes G the conditional probabilities $p(D|G)$ of observing such an alignment D. Where necessary, do not forget to consider both the base qualities and the mapping qualities.

(b) Use the approach of MAQ to determine the most likely genotype for this (potential) variant. Is it $\langle ref, ref\rangle$, $\langle ref, alt\rangle$ or $\langle alt, alt\rangle$?
Note: here, ref indicates the reference bases and alt the variant base; in the lecture we used the notation $\langle a, a\rangle$, $\langle a, b\rangle$ and $\langle b, b\rangle$.

(c) Calculate the posterior probabilities for all three genotypes, if you haven't done this already for (b). For the most likely genotype, determine the probability of a wrong genotype call and express it as a Phred score.

4. **Structural variants**

(Note: we haven't see such a figure in the lectures, so you won't need to interpret one in the exam ... :-)



(a) Which of the above figures depict the following structural variants (assign A-F)? Decide whether they (usually) are balanced or unbalanced.

Structural variant	**Figure**	**Balanced?**
Translocation		
Interspersed duplication		
Inversion		
Tandem duplication		
Deletion		
Novel sequence insertion		

5. **EWT procedure**

(Note: of course you won't be seeing such a complexity in the exam, because you won't have a computer or calculator at hand ... :-)

Given a chromosome with a length of 1 million bp and a window size of 100 bp. Let the desired false positive rate be 5%.

(a) What is the upper-tail probability threshold (EWT score) to detect a duplication of a segment of 1000 bases length?

(b) What is the corresponding minimum z-score that can be accepted for the windows?

Hints:

i) The z-score is normally distributed with a standard deviation of 1 around a mean of 0.

ii) In R, the inverse of `pnorm` is `qnorm`.